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ORIGINAL ARTICLE

Partial discharge detection of insulated conductors based on CNN-LSTM of attention mechanisms

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Abstract
Under the condition of a strong electric field, partial discharge often occurs when insulated wire is damaged. The recognition of partial discharge is an effective method for the fast and accurate detection of high voltage insulated wire faults. This paper proposes a PD recognition algorithm based on a convolutional neural network and long short-term memory (LSTM). In addition, attention mechanisms are introduced to give separate weights to LSTM hidden states through a mapping, weighting, and learning parameter matrix. This is done to reduce the loss of historical information and to strengthen the influence of important information. The complex relationship between the voltage signal change and the grid operation state response has been established. The proposed method is verified by the ENET data set published by VSB University. The recognition accuracy is 95.16% for no-PD and 94.44% for PD. Results from the proposed algorithm show that this method has a higher detection accuracy.

Keywords Partial discharge · Convolutional neural network · Long short-term memory · Attention mechanism · Fast Fourier transform

Abbreviations

PD	Partial discharge
IOC	Insulated overhead conductors
CNN	Convolutional neural network
LSTM	Long short-term memory
SVM	Support vector machines
ANN	Artificial neural network
MLP	Multi-layer perceptron
RNN	Recurrent neural network
FFT	Fast Fourier transform
K	Convolution Kernel
ϕ	Activation function
F	Frobenius norm
Mt	Intermediate variable
Ct	Dynamic feature vector
N	Number of samples
$\hat{y}^{(i)}$	Predicted on time i value
FNN	Fully convolutional neural network
BiLSTM	Bi-directional long short-term memory

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1 Introduction

Power equipment often has different degrees of failure during operation [1–3]. The insulation problem of power equipment is one of the main causes of power equipment failures and accidents. Through partial discharge monitoring, the insulation status of power equipment can be judged in advance. Insulated conductors can improve the stability of power transmission and reduce the covered area of power transmission equipment better than traditional bare conductors [4, 5]. Therefore, insulated conductors are increasingly used in overhead power transmission [6]. However, the use of insulated overhead conductors (IOC) faces major challenges. When an IOC is damaged from falling to the ground or from branches touching it, there can be a power outage or even a fire [7]. This kind of fault usually does not cause

overcurrent. Therefore, phase-to-ground and phase-to-phase faults usually cannot be distinguished by normal protective equipment [8]. However, such faults can cause partial discharge phenomena [9]. The current value of a PD is very small (about 6–10 A) and it can be easily interfered with by external background noise, which is the main problem when it comes to achieving accurate detection.

The most typical, common, and direct phenomenon of PD is the movement of charges between electrodes [10–12]. In the past, PD detection technologies mainly included electrical detection methods (such as ultra-high frequency detection methods) and non-electricity detection methods (such as infrared thermal, acoustic, and optical measurements). In recent years, the development of artificial intelligence has resulted in new ideas and new methods for identifying PD activities. The current methods for PD activity detection are mainly divided into traditional machine learning methods and deep learning methods. Traditional machine learning methods include support vector machines (SVM) [13], artificial neural networks (ANNs) [14], etc. A SVM can handle nonlinear problems. However, it is awkward when it comes to handling large-scale data. An ANN has self-learning and self-adaptive capabilities, and can fully solve the nonlinear problem in large-scale data. Thus, it has been widely used in the field of detection. Typical ANN methods include the multilayer perceptron (MLP) [15], wavelet neural network (WNN) [16], and BP neural network (BPNN) [17]. However, study has found that it is problematic to learn the correlation of time series data. The voltage data of an IOC has the characteristics of non-linearity, unevenness, and dynamic changes. As a result, the output of the model is related to both the current input and the past input [18–20].

In recent years, deep learning has been widely used. Deep leaning includes the deep neural network (DNN), automatic encoder (AE), convolutional neural network (CNN), recurrent neural network (RNN), long and short-term memory neural network (LSTM), etc. Accuracy is improved when a DNN is applied to PD activity detection, while the time sequence characteristics of voltage data are ignored in feature extraction [21]. AE belongs to unsupervised learning, which is mainly used for data dimensionality reduction and feature extraction. It usually needs to be combined with other models to be applied to the field of PD activity detection [22]. CNN includes convolution calculations and has a deep structure [23]. It uses convolution pooling to extract data features, which reduces the errors caused by artificial feature extraction [24]. The CNN method has a higher PD activity recognition accuracy than SVM and BPNN. However, when the voltage data is volatile and unstable, a single CNN model has a hard time learning the dynamic changes of voltage data [25]. RNN introduces a loop structure into the network, which helps it better to process dynamic time series data when compared to other neural networks [26]. LSTM is a special RNN that was proposed to solve the problem of gradient disappearances in RNN. It considers the dependence of long-term and short-term data information in the time series relationship [27]. RNN and LSTM are applied to PD recognition tasks. The low-level features of PD phase signals are taken as input [28]. However, these networks are prone to losing sequence information when the input time series is lengthy [29, 30]. The attention mechanism can assign distinct weights to the input features so that important information does not disappear with an increase of the step size. This can highlight the influence of more significant information, and make it easier to learn in a long-distance interdependent relationship [31].

The contributions of this paper are summarized as follows.

(1) Aiming at the problem of the low signal-to-noise ratio of data, FFT is used to preprocess data. The different threshold coefficients are compared and tested to determine the best coefficient.
(2) The CNN structure of the one-dimensional convolution layer and the pooling layer are constructed. The multi-dimensional features that reflect the complex dynamic changes of the voltage are extracted. The extracted feature vectors constructed in the form of a time series are used as the input of the LSTM network.
(3) It is difficult to effectively extract the potential high-dimensional features in the historical sequence, and important information is easily lost when the time sequence is too long. An LSTM neural network model is introduced that can enhance the capability of extracting time series information.
(4) An attention mechanism is introduced to give different weights to the LSTM hidden layer by mapping the weights and learning parameter matrix. This reduces the loss of historical information and enhances the influence of important information.

2 PD detection model

The model structure of PD detection is shown is Fig. 1, which mainly includes a noise reduction layer, an input layer, a CNN layer, a LSTM layer, an attention layer and an output layer. The noise of sequence voltage data is reduced by a FFT algorithm. The separate thresholds are analyzed to select the best threshold. Voltage signals of the noise reduction are used as the input of the CNN layer. The CNN layer is used to complete the feature vector extraction. The LSTM layer and the attention layer learn the internal change rules of the voltage signals from the extracted features, and complete the detection of PD activities. Finally, detection results can be obtained through the output layer.

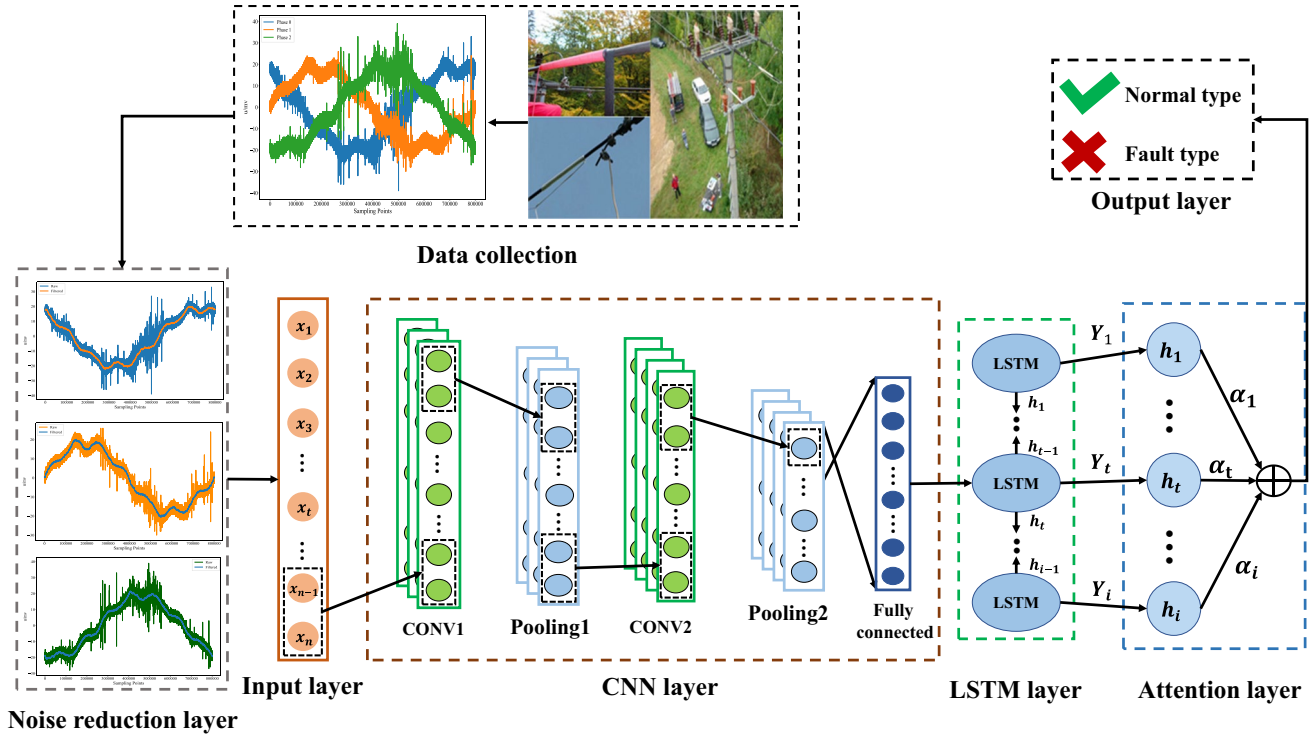


Fig. 1 CNN-LSTM model structure based on an attention mechanism

2.1 Convolutional neural network

A deep convolution neural network is made up by hierarchically arranged trainable layers that can learn effective features [32, 33]. A typical convolutional layer contains the input I and the convolution kernel K. Two-dimensional convolution is defined as follows:

$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n I(m,n)K(i-m,j-n) \tag{1}$$

Since the voltage fluctuation signal is one-dimensional data, a one-dimensional convolution is performed for each layer. That is, $m = 1$ in (1). One-dimensional convolution can be calculated as (2):

$$C_{ij}^l = \phi\left(k_{n\times 1}^j * x_{i:i+n}^i + b_{ij}\right) \tag{2}$$

where $k_{n\times 1}^j$ is the j-th kernel of the $n \times 1 \times j$ size, which belongs to the l-th convolutional layer. $x_{i:i+n}^i$ represents the i-th input. b_{ij} is the corresponding bias vector. ϕ is the activation function, which can be *Sigmoid*, *Tanh*, *Relu*, *Lacky Relu*, etc. C_{ij}^l is the i-th feature point of the j-th kernel in the l-th convolutional layer.

For classification problems, the fully connected layer should be connected to the last convolutional layer of the output classification result, which is defined as:

$$y^l = \phi\left(W^l y^{l-1} + b^l\right) \tag{3}$$

where W^l is the weight matrix connecting the upper layer and the current layer, y^{l-1} is the feature matrix of the upper layer, and b^l is the bias vector of the current layer. All of the parameters are updated by the back propagation method [34] to minimize the mistakes between the actual output y and the expected output $\hat{y}$. The loss function is defined as follows:

$$L = \frac{1}{2n}\|y - \hat{y}\|_F^2 \tag{4}$$

where F is the *Frobenius* norm.

2.2 Recurrent neural network

Unlike FNN and CNN, RNN can be regarded as a network with memory. It saves information from the previous moment as input for the next moment. Therefore, when compared with FNN and CNN, RNN can extract the features of time series data. The standard RNN includes 3 layers: input layer, hidden layer, and output layer, as shown in Fig. 2.where x_t represents the input vector at time t. In the hidden layer, each unit has an activation function. In each step, the RNN unit outputs a feature value h_t, which is used as the input of the next unit. The function of RNN can be shown as formula (5) and formula (6).

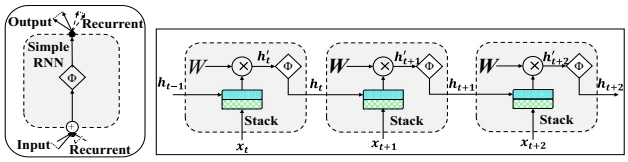


Fig. 2 Standard RNN structure

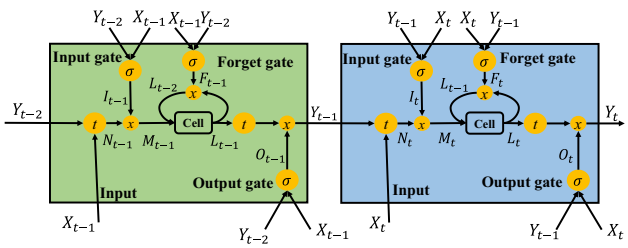


Fig. 3 Internal structure of LSTM

$$h_t^{'} = W \times i_t + R \times h_{t-1} + b \tag{5}$$

$$h_t = sigmoid\left(h_t^{'}\right) \tag{6}$$

where $h_t^{'}$ represents the hidden state at time t. In addition, W, R represent the weight matrix and b represents the bias vector, respectively. h_t is the output, and *sigmoid* is the activation function. Due to this special network structure, RNN has a memory function, which can extract the characteristics of time series data.

2.3 Long short-term memory neural network

With an increase of the data dimension, the learning ability of RNN can decrease since the standard RNN only considers the influence of adjacent information units. If the previous information unit is too far from the current unit, the RNN network may forget the data information. Thus, its learning capability is reduced. It is possible to obtain LSTM by improving RNN, which is good at extracting long-term features. When compared with the RNN neural network, LSTM can solve the problem of the vanishing gradient. In addition, LSTM is excellent in extracting features from multi-dimensional temporal correlation data [35–37]. To solve the problem of RNN, it adds three special gates, namely an input gate, a forget gate, and an output gate. The internal structure of LSTM is shown as Fig. 3.

The three gates are calculated by (7)–(9).

$$I_t = \sigma\left(W_i \times X_t + Z_i \times Y_{t-1} + b_i\right) \tag{7}$$

$$F_t = \sigma\left(W_f \times X_t + Z_f \times Y_{t-1} + b_f\right) \tag{8}$$

$$O_t = \sigma\left(W_o \times X_t + Z_o \times Y_{t-1} + b_o\right) \tag{9}$$

The activation function σ is a Sigmoid activation function. W_i, W_f, W_o, Z_i, Z_f, and Z_o are the weight matrixes of each gate. b_i, b_f, and b_o are the corresponding bias vectors. The input data are shown as (10).

$$N_t = th\left(W_t \times X_t + Z_t \times Y_{t-1} + b_t\right) \tag{10}$$

where *th* is the *tanh* activation function, and W_t and Z_t are the weight matrices. b_t is the input bias vector. After obtaining the three gate states and input information, the intermediate variable M_t can be described as (11).

$$M_t = I_t \times N_t \tag{11}$$

The state value of the input gate is between 0 and 1. The unit information L_t and the output state Y_t can be represented as (12) and (13), respectively. The state values F_t and O_t of the forget gate and the output gate are between 0 and 1.

$$L_t = M_t + F_t \times L_{t-1} \tag{12}$$

$$Y_t = th\left(L_t\right) \times O_t \tag{13}$$

2.4 Attention mechanism

However, when the data length increases, it is difficult to perform effective feature extraction. To improve the learning performance of LSTM, an attention mechanism is added [37, 38]. The output information of LSTM is used as the input of the attention mechanism. The attention mechanism multiplies the extracted time series feature vectors by weight coefficients. Then, it attains the best combination of weight coefficients. Its goal is to convert the fixed output y_t into a dynamic feature vector C_t. The calculation mainly includes three steps.

(1) Calculate the parameters of $u^{i,t}$ within the i-th time, as (14):

$$\begin{aligned} u^{i,t} &= V^T \times \tanh\left(W_a \times Y_t + b_a\right) \\ i, t &= 1, 2, \ldots, n_steps \end{aligned} \tag{14}$$

where $u^{i,t}$ is the score model of the matching degree between the input at the i-th time and the output at the t-th time. V^T, W_a, and b_a are parameters to be trained, and *tanh* is the activation function.

(2) Normalize the data obtained in the first step. Then, it is possible to obtain the weight coefficient $\alpha_{i,t}$ of each state, as (15):

$$\alpha_{i,t} = \frac{e^{u_{i,k}}}{\sum_{k=1}^{n_steps} e^{u_{i,k}}} \quad i,t = 1,2,...,n_steps \tag{15}$$

(3) The dynamic feature vector C_t is obtained by multiplying the output of LSTM by the probability, as (16):

$$C_t = \sum_{t=1}^{n_steps} \alpha_{i,t} \cdot Y_t \quad i,t = 1,2,...,n_steps \tag{16}$$

After the dynamic feature vector C_t is obtained, the decoding process resembles the method of using LSTM.

2.5 Optimizer and loss function

Adam (adaptive moment estimation) optimization algorithms are selected to optimize the model parameters in the training process. Adam is a first-order optimization algorithm that can substitute for the traditional stochastic gradient descent process. It can iteratively renew the weight of the neural network based on training data. As a result, the output value of the loss function reaches the optimal value. The loss function uses the SoftMax logarithmic cross entropy loss function, as (17).

$$L(\hat{y}, y) = -\sum_{i=1}^{N}\sum_{j=1}^{C} y^{(j)} \log\left(\hat{y}^{(j)}\right) \tag{17}$$

where N is the number of samples. $y^{(i)}$ is the actual value, which is a C-dimensional vector after one-hot encoding. $\hat{y}^{(i)}$ is the predicted value at the time i.

3 Experimental setup

3.1 Principle of partial discharge

The high-voltage levels for general transmission lines are 10 kV, 35 kV, 220 kV, etc. Due to the installation method and manufacturing processes, there may be some impurities in cable insulation material. This makes the electric field on the surface or internal areas of cable insulation material uneven.

When the air gap inside an insulator discharges, the gas in the air gap is freed and the neutral molecules are separated into positive and negative ions. These charges create a modern electric field in the opposite direction to the outside. The electric field strength in the air gap is $E_c = E_o - E_i$, as shown in Fig. 4.

The PD of high-voltage cables is a relatively complex physical process. It requires a variety of feature parameters to fully describe its state. The main parameters include the apparent discharge charges, discharge repetition rate, discharge energy, discharge average current, and discharge power. Long-term partial discharge causes serious damage to the insulation of the cable, which is mainly manifested in local heating, chemical active products, charged particle affects, and radiation.

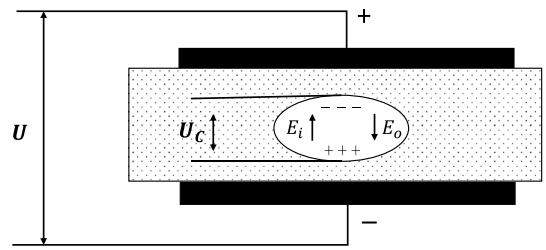


Fig. 4 Partial discharge generation mechanism

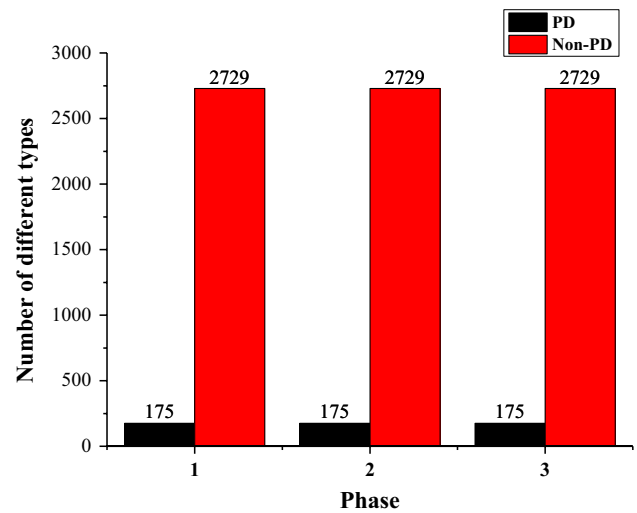


Fig. 5 ENET data set data distribution

3.2 Data set introduction

In 2018, VSB released the ENET data set on Kaggle, which is the world's largest data science collaboration platform. The data set consists of 8711 labeled voltage signals from four disparate locations. Those locations are on behalf of deployment in real environments (forested and difficult to access terrain). Each of the signals is a voltage waveform of 50 Hz, which consists of 800,000 data points and is pre-marked as PD (525) or non-PD (8186). The quantity distribution of disparate types of data is demonstrated in Fig. 5. Due to the large volume of the data set, the Hadoop distributed file system (HDFS) storage format is used. Examples of the PD and non-PD signals are shown in Fig. 6. What can be seen is that the maximum and minimum values of the non-PD signals are about 40 mV and − 40 mV, and that the fluctuation is relatively stable. On the other hand, the maximum and minimum values of the PD

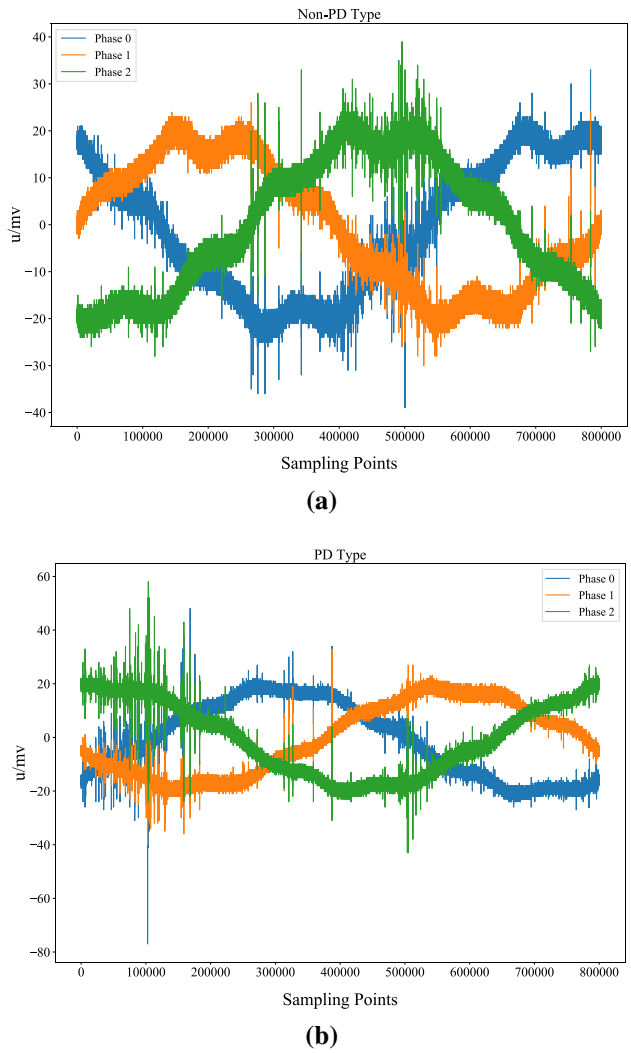


Fig. 6 Signal examples from ENET dataset: **a** normal type; **b** PD type

signal are about 60 mV and − 80 mV, and the signal fluctuation has markedly increased.

3.3 Data preprocessing

To facilitate the training of the model, the min–max method is used to normalize the original data to (− 1,1), as (18).

$$x_n = \frac{x - x_{\text{mean}}}{x_{\max} - x_{\min}} \tag{18}$$

where x is the original voltage. x_{mean}, $x_{\max}$, and $x_{\min}$ are the average, maximum, and minimum values of the voltage sample data.

3.4 Data noise reduction

In view of the noise of the original data (environmental noise, acquisition device noise, etc.), FFT noise reduction algorithms are used to preprocess the original data.

The Fourier transform of the one-dimensional signal x is:

$$f_j = \sum_{k=0}^{n-1} x_k e^{\frac{2\pi i}{n} jk}, \forall j = 0, ..., n-1 \tag{19}$$

Formula (19) is used to process the voltage signal. The frequency coefficients take the following values:

If n is even:

$$f = \frac{1}{dn}\left[0, 1, ..., \frac{n}{2} - 1, -\frac{n}{2}, ..., -1\right] \tag{20}$$

If n is odd:

$$f = \frac{1}{dn}\left[0, 1, ..., \frac{n-1}{2}, -\frac{n-1}{2}, ..., -1\right] \tag{21}$$

The frequency coefficient thresholds are set, and experimental results are shown in Fig. 7. These results show that when the threshold is 10^3, a lot of detailed information is ignored. In addition, when the threshold is set to 10^7, the signal after noise reduction still has a lot of noise information. Thus, the threshold is set to 10^5, and the noise reduction effect is the best.

3.5 Evaluation indicators

The accuracy, recall, *F1* value, and accuracy are chosen as evaluation indicators to comprehensively assess the performance of the model. The four indicators are shown as (22)–(25). The labels of the real results and the forecast results are shown in Table 1.

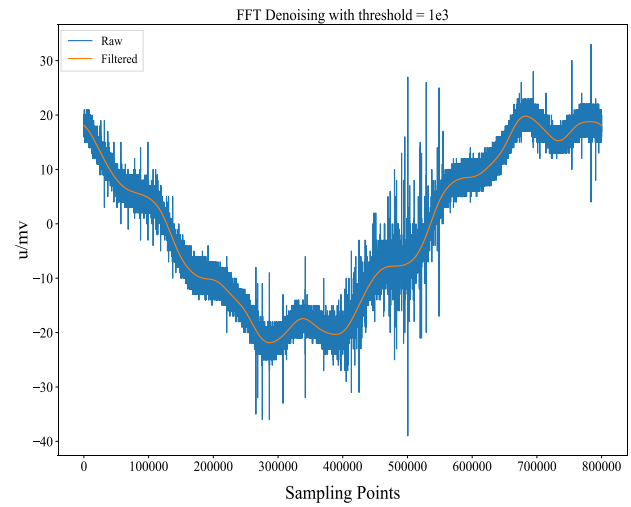
$$\text{precision } (\%) = \left(\frac{\text{TP}}{\text{TP} + \text{FP}}\right) \times 100\% \tag{22}$$

$$\text{recall } (\%) = \text{sensitivity } (\%) = \left(\frac{\text{TP}}{\text{TP} + \text{FN}}\right) \times 100\% \tag{23}$$

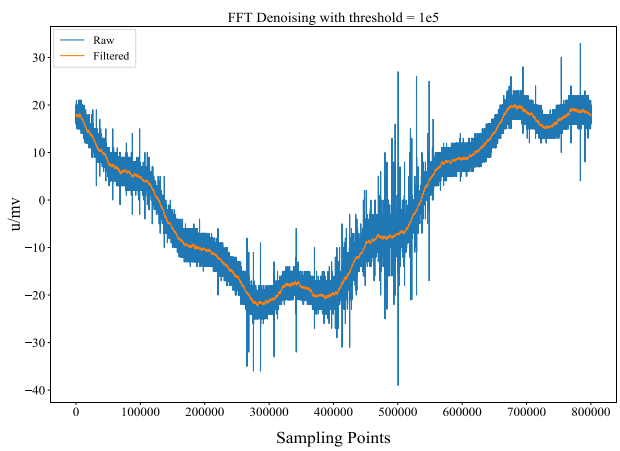
$$F1(\%) = \left(2 \times \frac{\text{prec} \times \text{rec}}{\text{prec} + \text{rec}}\right) \times 100\% \tag{24}$$

Table 1 True result and forecast result labels

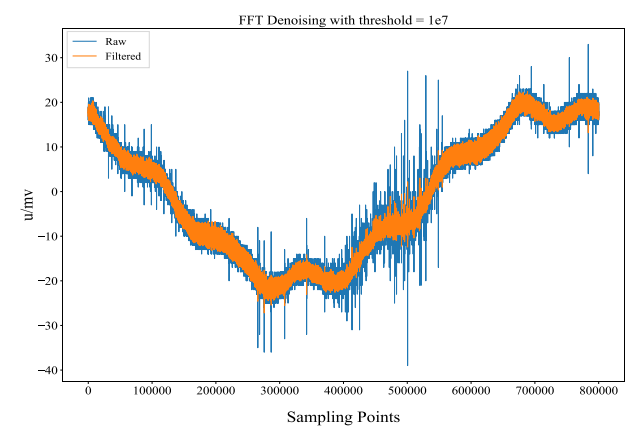
True results	Forecast results	
	Positive	Negative
Positive	TP	FN
Negative	FP	TN



(a)



(b)



(c)

Fig. 7 Data noise reduction comparison experiment: **a** threshold = 1 X 10^3; **b** threshold = 1 X 10^5; **c** threshold = 1 X 10^7

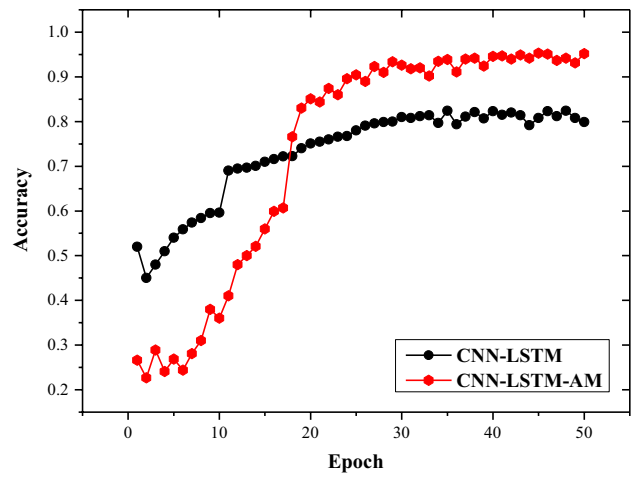


Fig. 8 Accuracy comparison between CNN-LSTM and CNN-LSTM-AM

$$\text{accuracy}\ (\%) = \left(\frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}\right) \times 100\% \tag{25}$$

4 Algorithm comparison and analysis

4.1 Comparison of CNN-LSTM and CNN-LSTM-AM

To confirm the detection model performance after adding an attention mechanism, comparative experiments are carried out on CNN-LSTM and CNN-LSTM-AM. The experimental results are shown in Fig. 8.

It can be seen that the convergence speed can be improved by combining CNN-LSTM-AM. In the early stages of training, the accuracy of CNN-LSTM-AM barely changes, while that of CNN-LSTM slowly increases. Since the attention mechanism can retain more of the characteristics of the time series data, when the network attains the optimal gradient descent direction, the accuracy of CNN-LSTM-AM rapidly increases. The accuracy of CNN-LSTM-AM is higher than that of CNN-LSTM.

4.2 Comparison with other algorithms

The proposed method is compared with typical PD detection algorithms such as the multilayer perceptron (MLP), support vector machine (SVM), fully convolutional neural network (FNN), CNN, BiLSTM, and the accuracy scores are used as an evaluation index. 6500 samples are randomly chosen as the training set and 2000 samples are chosen as the test set in the data set. In Fig. 9, the T-SNE algorithm is used to

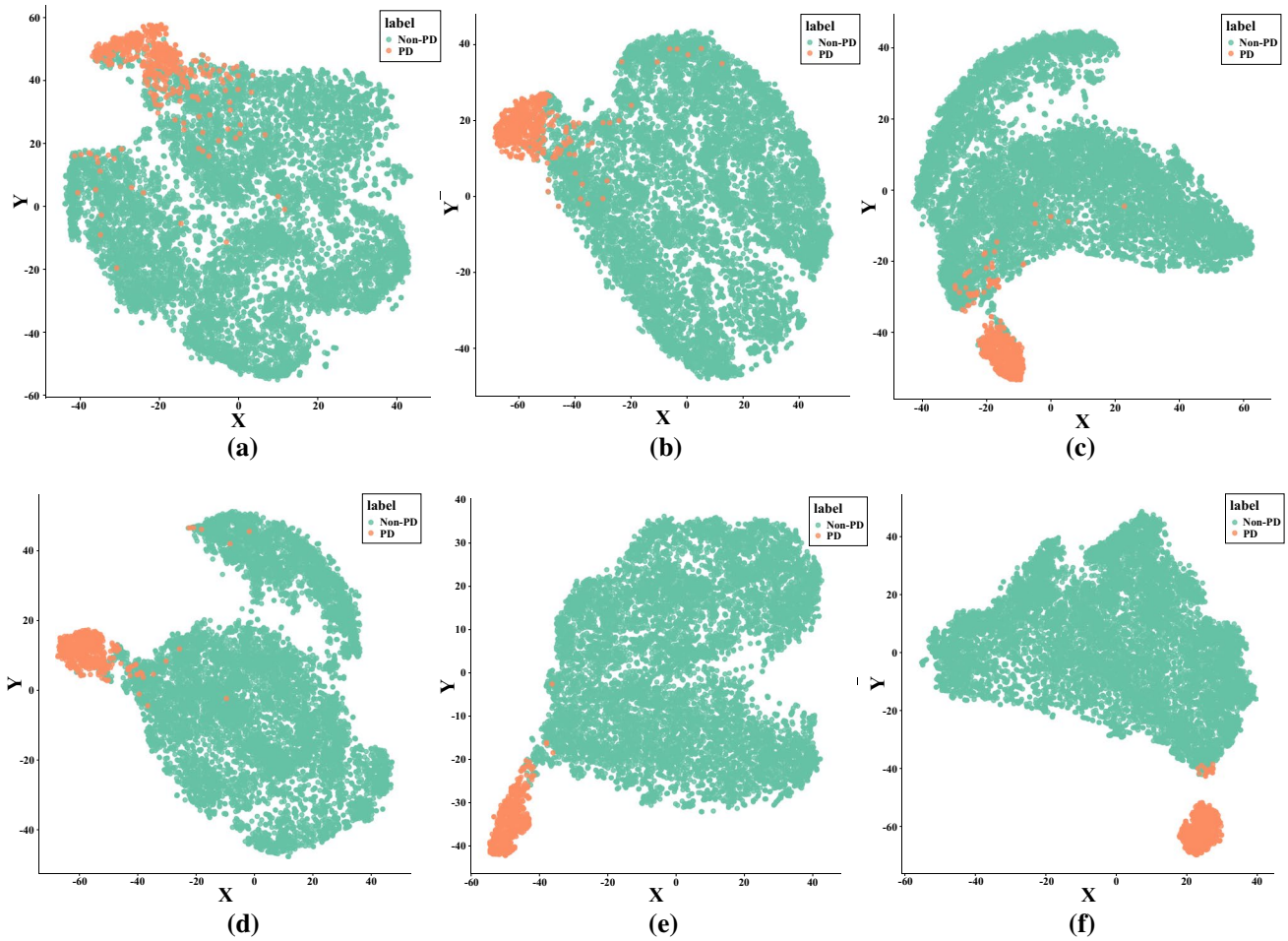


Fig. 9 Visualization of classification results: **a** MLP; **b** SVM; c FNN; **d** CNN; **e** BiLSTM; **f** CNN-LSTM-AM

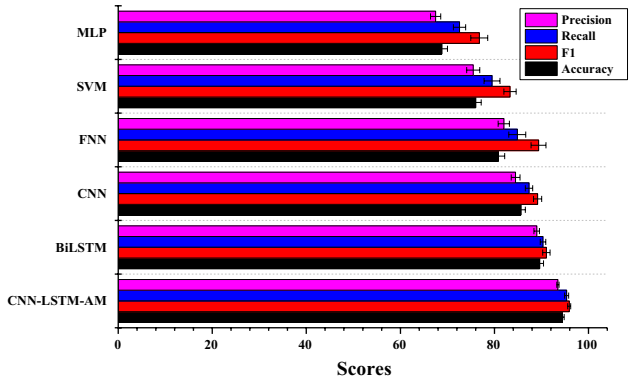


Fig. 10 Accuracy comparison of various algorithms

visualize the classification results on the ENET data set. It shows that the classification accuracy of traditional machine learning models is worse than that of deep learning models since the feature extraction capabilities of the traditional machine learning models are limited.

Figure 10 shows the evaluation index scores of distinct algorithms. It can be seen that traditional machine learning models such as MLP and SVM have difficulty extracting the deep feature information of voltage data. In addition, the classification accuracy of PD activity detection is unsatisfactory.

Deep learning models such as FNN, CNN, and BiLSTM can effectively extract deep feature information, which improves the detection accuracy to a certain extent. However, since a voltage data sequence contains 800,000 points, which means the data length is a long sequence, the above networks are prone to losing sequence information. Thus, their classification accuracy is less than 90%. The four parameter indicators of the proposed method are greater than 93%, which is why it outperforms the other methods. The accuracy of different algorithms is shown in Table 2.

Table 2 Accuracies of different algorithms

	Non-PD	PD	Accuracy average
MLP	68.8%	65.3%	67.05%
SVM	76%	74.67%	75.34%
FNN	80.8%	84%	82.4%
CNN	85.6%	82.67%	84.14%
BiLSTM	89.6%	88%	88.8%
CNN-LSTM-AM	94.44%	93.33%	93.89%

5 Conclusion

This paper proposes a PD activity detection method based on the CNN-LSTM-AM model. In addition, it reaches the following conclusions.

(1) Aiming at the problem where there is a lot of background noise in the voltage data, a FFT algorithm is used to analyze the noise reduction. Different thresholds are set for comparative experiments. The best transform threshold is used for the experimental parameters. The results show that the FFT method has an excellent noise reduction effect.

(2) CNN is used to complete the automatic feature extraction of the voltage data sequence. The CNN model has a stronger capability of nonlinear feature learning and fault feature analysis. This lays the foundation of further fault classification.

(3) Combined with an attention mechanism, the LSTM network is used to build the PD detection model. When compared with MLP, FNN, SVM, CNN, BILSTM, and CNN-LSTM models, the proposed detection model can achieve higher accuracy. This shows that the proposed algorithm has excellent PD detection performance and a wider application.

In the future, our main research work will be as follows. First, different types of data are often unbalanced. For example, there is a lot of data in the first type, while there is little data in the second type. In order to boost detection accuracy, the imbalance problem among different types of data sets will be studied. Second, the detection method of multi-algorithm combinations will be studied to improve the universality of the PD activity detection model, the ability to process large-scale data, and the practical application value of the proposed model.

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